

Thomas Rolf Mannhart

Professional AI Engineer

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SUMMARY

I'm a Professional AI Engineer at bbv Software Services in Zürich, building enterprise AI on the bbv AI Hub. I studied Informatics at the University of Zürich (BSc + MSc, AI specialization) and have been shipping software professionally since 2019 — from biomedical Java apps to full-stack web platforms to LLM-powered agentic systems.

EXPERIENCE

2025 — present

Professional AI Engineer — bbv Software Services AG, Zürich

Development of a comprehensive enterprise AI platform, including architecture design in collaboration with the software architect. Architecture and implementation of customized AI solutions — especially RAG systems — for customer projects in industry and market research. Technical leadership (Dev Lead) of customer AI projects. Operation and maintenance of the AI platform and customer-specific solutions. Consulting customers on their IT and AI strategy.

Python, LLMs / RAG, Agentic AI, Dev Lead, AI Strategy, Azure, TypeScript, Platform Engineering

2023 — 2024

Professional Software Engineer — Ergon Informatik AG, Zürich

Developed a time-tracking and workforce planning system for the retail sector. Handled end-to-end software delivery from requirements engineering and prototyping through to second/third-level support. Mentored new team members and contributed to youth outreach by organizing IT workshops for students.

Java, Kotlin, Angular, TypeScript, SQL, Selenium, Jenkins, Git

2020 — 2023

Senior Software Developer — PolygonSoftware, Opfikon

Led full-stack development of web applications and computer vision / machine learning projects at a UZH-founded startup. Designed software architectures, supervised dev teams, and interfaced directly with product owners and clients. Recognized for outstanding quality, reliability, and initiative.

Full Stack, Computer Vision, Machine Learning, Web Apps, DevOps

2019 — 2020

Junior Software Developer — swissbiomechanics ag, Zürich

Led an independent software project for this ETH spin-off, building a Java application to track biomedical analyses and automatically generate clinical reports. Handled stakeholder communication, requirements analysis, and coordinated with other developers.

Java, Report Generation, Biomedical

EDUCATION

2020 — 2023

MSc in Informatics (Artificial Intelligence) — University of Zürich

Thesis: KroneDB — Compressing and Querying Time Series Data using the Kronecker Decomposition. Supervised by Johannes Marti and Dan Olteanu at the Data Systems and Theory group.

2017 — 2020

BSc in Informatics (Software Systems) — University of Zürich

Thesis: A General-purpose Range Join Algorithm for PostgreSQL. Supervised by Michael Böhlen and Anton Dignös at the Database Technology group. UZH Semester Award 2020.

SKILLS

Programming: Python (expert), TypeScript / JS (advanced), Java / Kotlin (advanced), SQL (advanced)

AI: Agent Orchestration (expert), RAG (expert), MCP (advanced)

Tools: Claude Code (expert), Git / GitHub (expert), Docker (advanced)

Languages: German (native), English (fluent), French (fluent)

TALKS & PROJECTS

KI als Entwicklungspartner — bbv webinar on practical methods, tools, and proven strategies for integrating AI into the software development lifecycle — from requirements analysis through code development to testing.

AI-Augmented Software Engineering — Talk at the FHNW Data Science & Data Engineering Alumni Event on integrating AI into the software development lifecycle — from enterprise AI strategy and the bbv AI Hub, to working with coding agents and the "Development Funnel" for context-driven AI workflows.

About the bbv AI Hub — Swiss-made, model-agnostic enterprise AI platform — soon to be open-sourced. Listed on Siemens Xcelerator, certified Swiss Made Software.